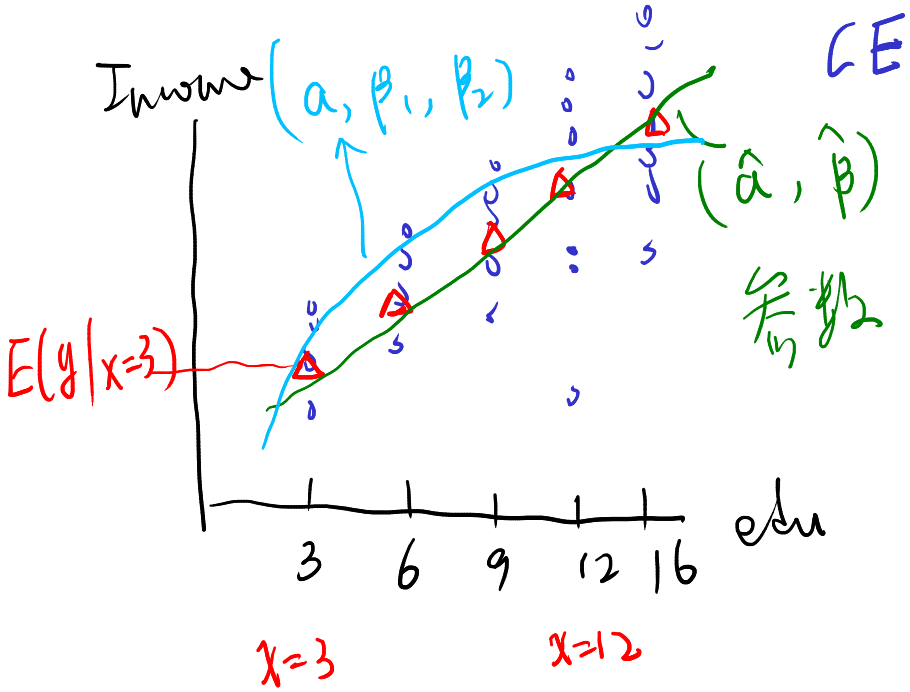


CEF

① $E(y|x) = m(x)$ 建模 $\longrightarrow$ $\begin{cases} E(y|x) = a + \beta \cdot x \\ E(y|x) = a + \beta_1 x + \beta_2 x^2 \end{cases}$

Data 分布 → 关系 $y \sim x$ (modeling)

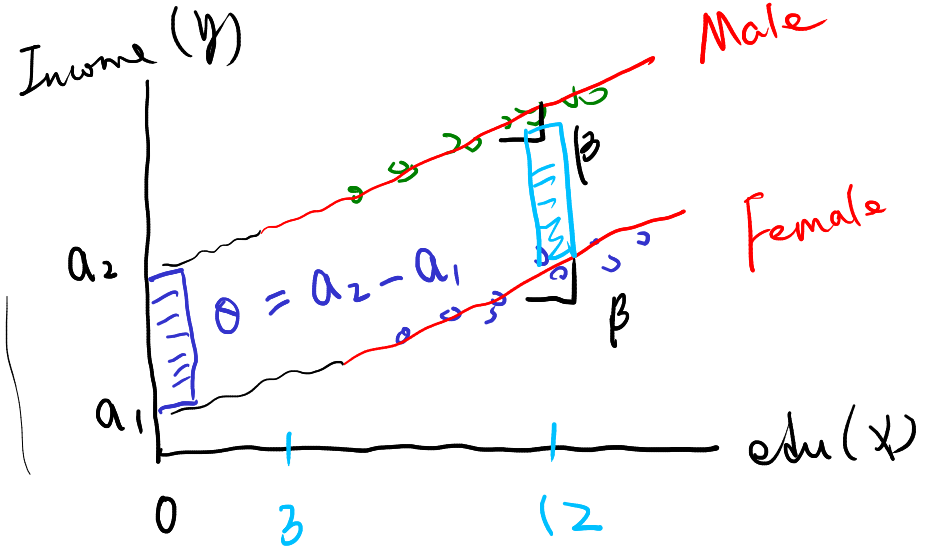
model → Data 联系起来

② $y_i = m(x_i) + \varepsilon_i$

个人 $E(y_i|x_i)$ 某个人收入

$\mathrm{cov}(\varepsilon_i, x_i) = 0$

$E(\varepsilon_i | x_i) = 0$



y	x	D
20.3		0
13.8		0
27.0		0
13.0		0
⋮		1
		1
		1

$$D = \begin{cases} 0 & \text{Female} \\ 1 & \text{Male} \end{cases}$$

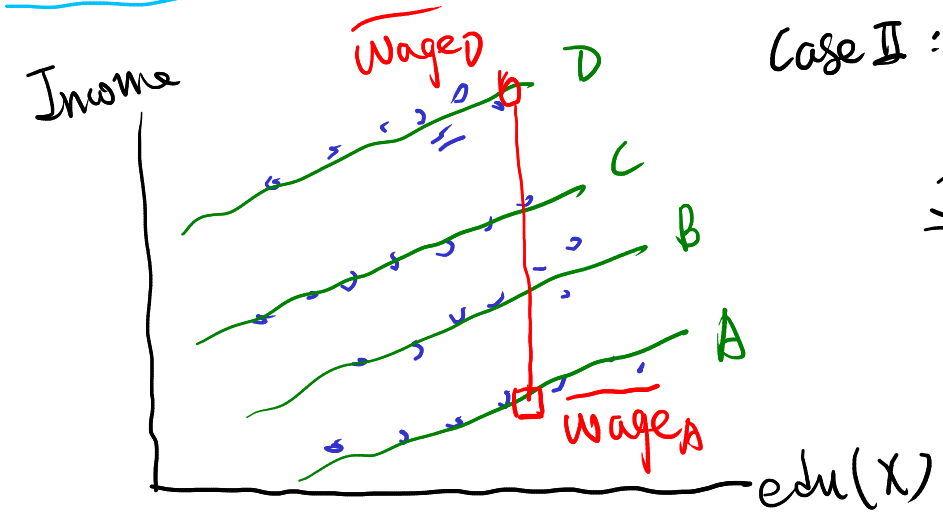
$$y_i = a_1 + D_i \cdot \theta + x_i \beta + \varepsilon_i$$

$D=0$ (F): $y_i = a_1 + x_i \beta + \varepsilon_i$

$D=1$ (M): $y_i = (a_1 + \theta) + x_i \beta + \varepsilon_i$

Case I: 只有一个 Dummy

Case II: 多个 Dummies



j	Industry	D_1	D_2	D_3	D_4	1
1	A	1	0	0	0	1
2	B	0	1	0	0	1
3	C	0	0	1	0	1
4	D	0	0	0	1	1

$$y = \cancel{a_1} + \theta_1 D_1 + \theta_2 D_2 + \theta_3 D_3 + \theta_4 D_4 + x_i \beta + \varepsilon_i$$

$$y_{ji} = a + \theta_j + x_{ji} \cdot \beta + \varepsilon_{ji}$$

Data

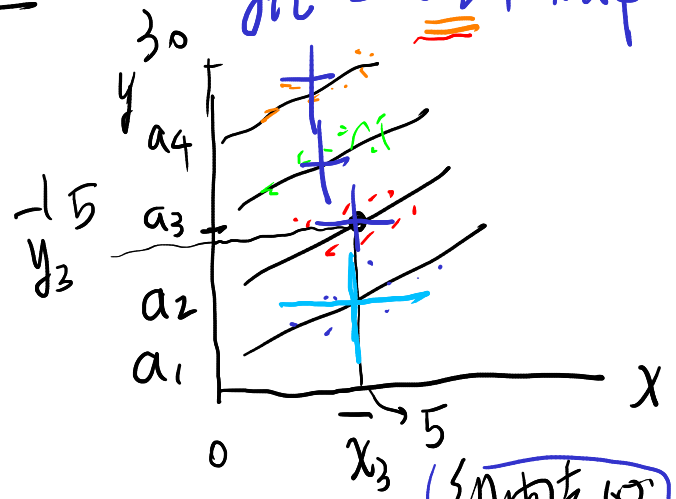
id	year	y	x
1	2021		
1	2022		
1	2023		
2	2021		
2	2022		
2	2023		
3	2022		
3	2023		

$N = 4000$ 家，$T = 20$ 年

↓

a_i（？每个系数）

$$y_{it} = a_i + x_{it}\beta + \varepsilon_{it} \quad ①$$



$$\bar{y}_i = a_i + \bar{x}_i\beta + \varepsilon_i \quad ②$$ （每家的均值）

$$\bar{y}_i = \frac{1}{3}(y_{i1} + y_{i2} + y_{i3})$$

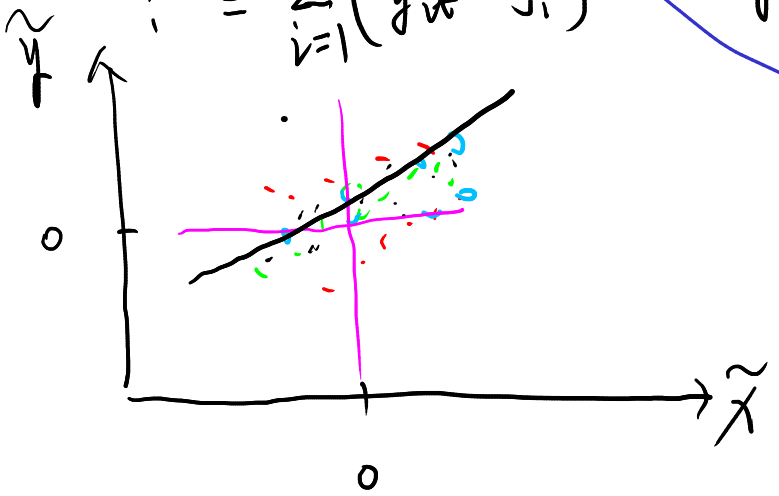
① − ②

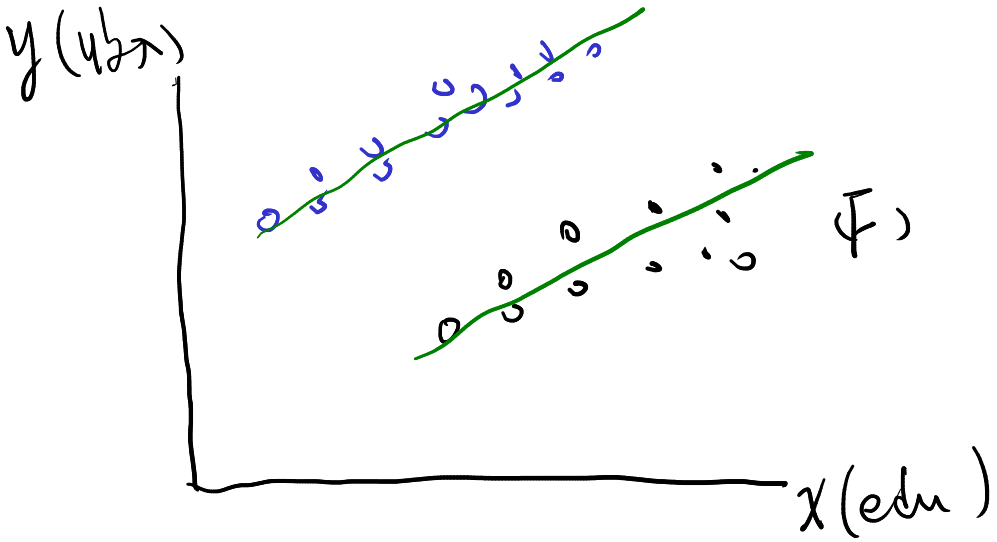
组内去心

$$\underbrace{(y_{it} - \bar{y}_i)}_{\tilde{y}} = \cancel{a_i} + \underbrace{(x_{it} - \bar{x}_i)}_{\tilde{x}}\beta + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

干掉 a_i

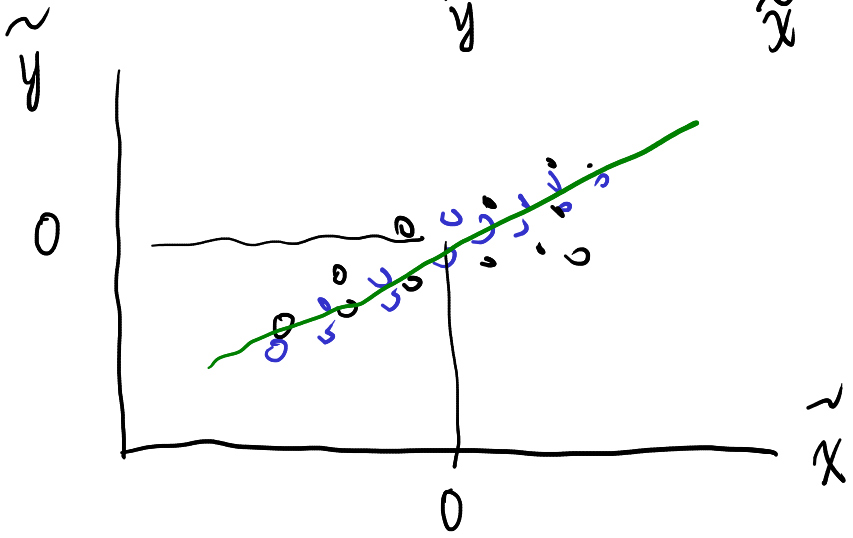
$$? = \sum_{i=1}^{N}(y_{it} - \bar{y}_i)$$

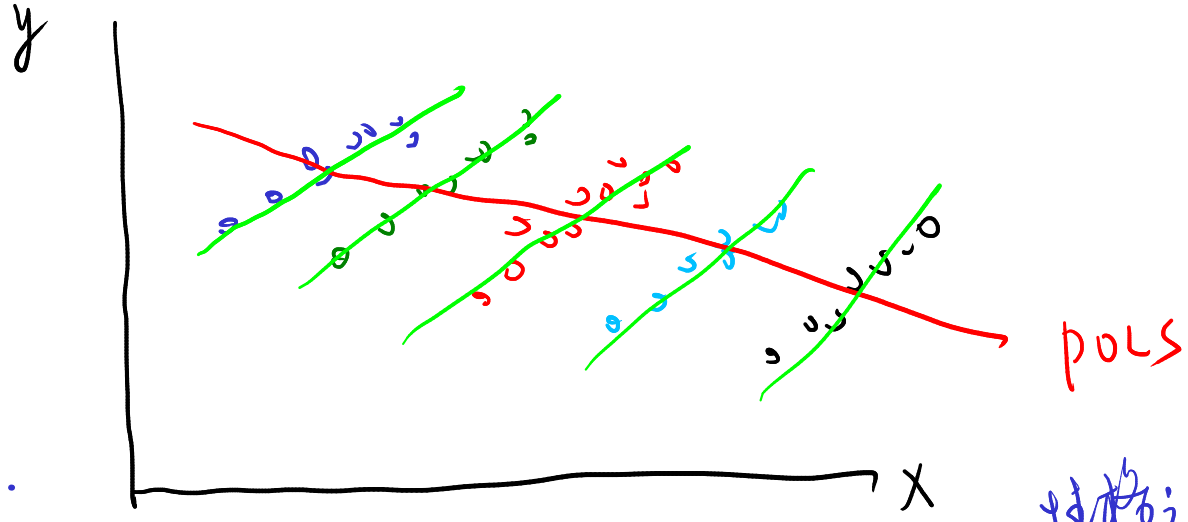




去掉 $\downarrow$ $(y_{it}-\bar{y}_i) \sim (x_{it}-\bar{x}_i)$

$\tilde{y}$ $\tilde{x}$





POLS: $y_{it} = a_0 + x_{it}\beta + \boxed{\varepsilon_{it}}$

$a_i + U_{it}$

个体效应

FE: $y_{it} = a_i + x_{it}\beta + U_{it}$

$$y = a + \underbrace{D\cdot\theta}_{X_1\beta_1} + \underbrace{m(x_{it})}_{X_2\beta_2} + \varepsilon_{it}$$

$D = \begin{cases} 0 & \text{Control 组 (未一)} \\ 1 & \text{Treat 组 (开立高铁)} \end{cases}$

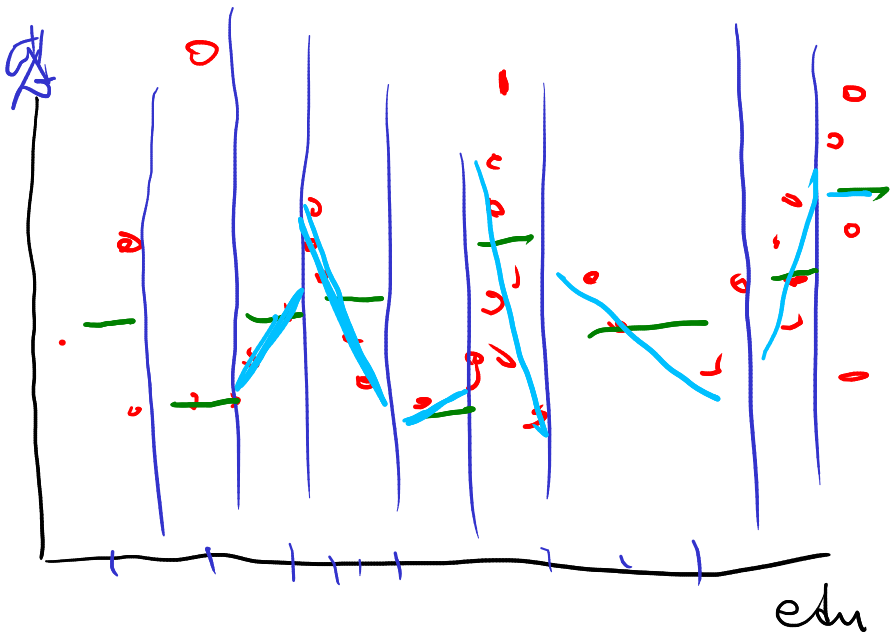
$D = g(x_{it})$ $m(\cdot)$ 非线性

DML (Double Machine learning) = 机器学习方法 $g(\cdot)$ CNN

Step1: fit y on x_{it} → $\hat{m}(x_{it})$

Step2: fit D on x_{it} → $\hat{g}(x_{it})$

Step3: $\underbrace{y_{it} - \hat{m}(x_{it})}_{\tilde{y}_{it}}$ on $\underbrace{D - \hat{g}(x_{it})}_{\tilde{D}_{it}}$

$\hat{\theta}$

幸福感